CLASS 9 — MATHS · FORMULA SHEET

Chapter 10: **How Quantities Combine (Understanding Data)** | Quick revision, one page

Averages

- Average $= \dfrac{\text{sum of values}}{\text{number of values}}$; sum = average $\times$ number of values.
- Combined average of two groups: $\dfrac{n_1 a_1 + n_2 a_2}{n_1 + n_2}$ (not $\dfrac{a_1 + a_2}{2}$ unless $n_1 = n_2$).
- The combined average always lies between the two group averages, nearer the average of the larger group.

Weighted average

- Weighted average $= \dfrac{w_1 x_1 + w_2 x_2 + \cdots}{w_1 + w_2 + \cdots} = \dfrac{\sum wx}{\sum w}$.
- Equal weights give the simple mean.

Mixtures and concentration

- Concentration $= \dfrac{\text{amount of ingredient}}{\text{total amount of mixture}} \times 100\%$.
- Amount of ingredient $= \dfrac{\text{concentration}}{100} \times \text{total}$.
- On mixing: total ingredient = sum of ingredients; total volume = sum of volumes; new concentration $= \dfrac{\text{total ingredient}}{\text{total volume}} \times 100\%$.
- Adding pure water keeps the amount of ingredient the same and increases the total.

Stacked bars

- A stacked bar chart shows the parts of each total on one bar; bar length = total.
- In a 100% stacked bar chart every bar has length 100% and shows only the shares of the parts.

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